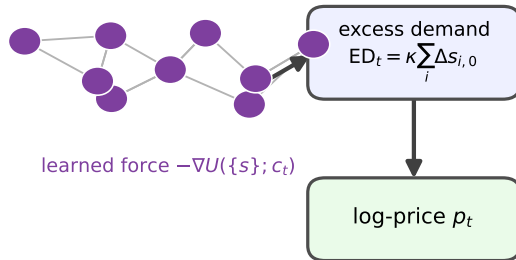


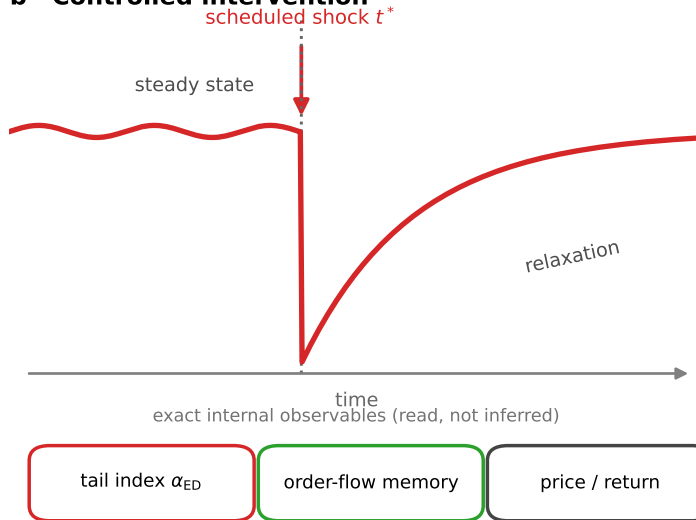
a Particle market model

N agents $s_i \in \mathbb{R}^d$



overdamped Langevin + MACE-lite force;
fully differentiable

b Controlled intervention



c Four controlled results

1 Measurement correction

warm-up inflates the tail

2 Driven transient

heavy tail only out of equilibrium

3 Mechanism + OFI signature

coordinated flow, not price shocks

4 Real-market boundary

no crash heavy-up $\Rightarrow$ OF prediction